

Adam Pikielny

apikielny.github.io / apikielny@gmail.com / Scholar / LinkedIn / GitHub / PDF

Research Scientist at Adobe.

Deep Learning · Computer Vision · Generative Models · Computational Photography.

I work on the full AI stack: from novel research through large-scale distributed and synthetic-data training to on-device deployment. I've authored a CVPR 2025 oral (top 0.7%) and three patents, and shipped production models used by millions.

Experience

Research Scientist, Adobe — Marc Levoy's computational photography group

2022–present

- Built Reflection Removal end-to-end on a team of 2, from our CVPR oral to production: a 2.5M-parameter model — 50× smaller than the prior 125M-parameter SOTA — trained purely on synthetic data yet outperforms prior SOTA by 4+ dB PSNR on real photographs.
- Optimized the model for on-device inference (CoreML/ONNX) — 1K-resolution preview in 4.5–6.5s on a MacBook or iPhone — and shipped it to millions of users in Lightroom, Photoshop, Camera Raw, and Project Indigo.
- Contributed to Project Indigo, mentored research interns, and authored 2 publications and 3 patents.

Research Intern, Adobe

Summer 2020 and 2021

- Built After Effects' first ML-based effect plugin, leading to a patent.

Publications

Hist2Style: Histogram-Guided Stylization with Bilateral Grids

CVPR 2026

Dekel Galor, **Adam Pikielny**, Zhoutong Zhang, Ke Wang, Laura Waller, Jiawen Chen, Ilya Chugunov. Also spotlight at the CCD Workshop.

Removing Reflections from RAW Photos

CVPR 2025

Eric Kee, **Adam Pikielny**, Kevin Blackburn-Matzen, Marc Levoy. **Oral, top 0.7% of submissions.**

Consistent Depth Estimation for Video

Brown CS Senior Thesis, 2022

Adam Pikielny, Marc Mapeke, James Tompkin. **1st Prize, Brown CS Undergraduate Research Symposium.**

Towards Social Video Verification of Face Expression, Identity, and Lighting for DeepFake Detection

Preprint, 2021

Alex Meyerowitz*, **Adam Pikielny***, Harman Suri, Eleanor Tursman, James Tompkin. *equal contribution.

Education

Brown University, B.S. Computer Science, Visual Computing and AI/ML pathways

2022

- Senior thesis, *Consistent Depth Estimation for Video*, won 1st Prize at the Brown CS Undergraduate Research Symposium.
- Graduated with Honors and won the Senior Prize in Computer Science.
- Research in James Tompkin's Visual Computing Lab on deepfake detection and depth estimation.

Skills

Tools: Python, C++, PyTorch, DDP, CoreML, ONNX, CUDA, Halide, Bazel

Models & architectures: generative models (diffusion & flow models), transformers, GANs, CNNs

Methods: synthetic data generation, distributed training, foundation-model distillation, adversarial training, image & signal processing, on-device optimization (quantization, palettization)

Shipped Products

Reflection Removal: Adobe Camera Raw (2024), Lightroom (2025), Photoshop (2026), Project Indigo (2025) 2024–2026

One-click removal of window reflections from photographs. Productized from my CVPR research and 2 patents.

Project Indigo, Adobe Nextcam 2025

iOS camera app with full manual controls, an SLR-like look, and the highest image quality on iPhone in both JPEG and RAW. Ranked #5 in top free video/photo apps in the US one week after launch and hit 1M downloads in four weeks with zero marketing.

Patents

Removing Image Overlays filed Jan 2024

Eric R. Kee, **Adam A. Pikielny**, Marc S. Levoy. US Patent Application 18/426,758. Adobe Inc.

User Interface For Adjusting Image Effects Depicted In Images filed Dec 2024

Thomas Knoll, Eric R. Kee, **Adam A. Pikielny**. US Patent Application 18/975,655. Adobe Inc.

Machine Learning-Based Chroma Keying Process Oct 2024

Seoung Wug Oh, Joon-Young Lee, Brian Price, John G. Nelson, Wujun Wang, **Adam Pikielny**. US Patent 12,106,398. Adobe Inc.

Awards & Honors

- CVPR 2025 Oral, *Removing Reflections from RAW Photos* (top 0.7% of submissions)
- Senior Prize in Computer Science, Brown CS, 2022
- 1st Prize, Brown CS Undergraduate Research Symposium, senior thesis, 2022
- Adobe MAX Sneaks 2023, *Project See Through*

Invited Talks

Guest lecture, *Reflection Removal*, CSCI 2240: Advanced Computer Graphics, Brown University Apr 2026

Guest lecture, *On-Device Computational Photography*, CSCI 1290: Computational Photography, Brown University Oct 2025

Teaching

Teaching Assistant, CSCI 0220 Discrete Structures and Probability, Brown CS

2020

Teaching Assistant, CSCI 1230 Introduction to Computer Graphics, Brown CS

2021

Open Source Contributions

pconfigs, Adobe (Apache 2.0)

A dataclass-based Python config library. Configs are typed, inheritable Python classes that live next to the code they configure, instead of brittle text files that drift out of sync. We used it throughout the reflection removal project.

Press

Project Indigo

The Verge / PetaPixel / DPReview / DPReview Gear of the Year / AppleInsider / Tom's Guide / Popular Science / Six Colors / Fstoppers / BGR

Reflection Removal

The Verge / Engadget / PetaPixel / DPReview / Digital Camera World / Creative Bloq / Fstoppers / TechRadar (MAX Sneaks 2023) / Brown CS

References available upon request.